

Ajin Cho

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Research Interests

Air-sea interaction under extreme conditions (e.g., tropical cyclones, strong ocean currents)
Coupled model simulations

Research Experience

Colorado School of Mines

Postdoctoral Researcher

Golden, CO, USA

Mar. 2026 – Present

Education

Yonsei University

Ph.D. in Atmospheric Sciences

Advisor: Dr. Hajoon Song

Seoul, Rep. of Korea

Sep. 2021 – Feb. 2026

Yonsei University

M.S. in Atmospheric Sciences

Advisor: Dr. Hajoon Song

Seoul, Rep. of Korea

Sep. 2019– Aug. 2021

Yonsei University

B.S. in Atmospheric Sciences

Seoul, Rep. of Korea

Mar. 2014 – Feb. 2019

Publications

Published.....

Cho, A., Song, H., Moon, I.-J., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., & Miller, A. J. (2026). Modulation of Tropical Cyclone Intensity by Current-Wind Interaction, *npj Climate and Atmospheric Science*, 9, 44. doi:10.1038/s41612-025-01316-1

Cho, A., Song, H., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., Miller, A. J. (2025). Dynamic and Thermodynamic coupling between the Atmosphere and Ocean near the Kuroshio Current and Extension System, *Ocean Modelling*, 194. doi:10.1016/j.ocemod.2024.102496

Lee, C., Song, H., Choi, Y., **Cho, A.**, & Marshall, J. (2025). Observed multi-decadal increase in the surface ocean's thermal inertia. *Nature Climate Change*, 15, 308-314. doi:10.1038/s41558-025-02245-w

Cho, A., Song, H., Tak, Y.-J., Yeh, S.-W., An, S.-I., Lee, S.-M., Ji, H.-S., Hyun, Y.-K. (2022). Atmosphere-driven Cold SST Biases over the Western North Pacific in the GloSea5 Seasonal Forecast System. *Climate Dynamics*, 59(9), 2571-2584. doi:10.21203/rs.3.rs-786427/v1

In preparation.....

Cho, A., Williams, A., Davis, J., & Villas Bôas, B. Relative Roles of Winds and Currents in Surface Wave

Teaching Experience

Advised 8-week research: guided analysis, literature review, and presentation <i>Undergraduate Mentoring</i>	Seoul, Rep. of Korea <i>Spring 2025</i>
Advised 8-week research: supported data, analysis, and presentation <i>Undergraduate Mentoring</i>	Seoul, Rep. of Korea <i>Spring 2022</i>
Air-Sea Interaction, taught by Prof. Hajoon Song at Yonsei Univ. <i>Teaching Assistant</i>	Seoul, Rep. of Korea <i>Spring 2020</i>

Awards & Scholarships

2024: Best Student Poster Award (Ocean Science Section), Asia Oceania Geosciences Society (AOGS)
2018: Honors with Excellent Grade (Spring Semester), Yonsei University

Invited Talk

Physical Oceanography Dissertation Symposium (PODS) <i>Impact of Sea Surface Currents on Air-Sea Interaction.</i>	Lihue, HI, USA <i>Dec. 2026 (Scheduled)</i>
Air-Sea Flux from Space Webinar Series <i>Current-Wind Interaction and Tropical Cyclones.</i>	Virtual Meeting <i>Feb. 2025</i>

Conferences

OSM 2026 (talk, poster) <i>Modulation of Tropical Cyclone Intensity by Current-Wind Interaction.</i> Cho, A., Song, H., Moon, I.-J., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., & Miller, A. J.	Glasgow, Scotland <i>Feb. 2026</i>
AOGS Annual Meeting (talk) <i>Modulation of Tropical Cyclone Intensity by Current-Wind Interaction.</i> Cho, A., Song, H., Moon, I.-J., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., & Miller, A. J.	Singapore <i>Aug. 2025</i>
BACO-25 (talk) <i>Modulation of Tropical Cyclone Intensity by Current-Wind Interaction.</i> Cho, A., Song, H., Moon, I.-J., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., & Miller, A. J.	Busan, Rep. of Korea <i>Jul. 2025</i>
KMS Spring Meeting (talk) <i>Modulation of Tropical Cyclone Intensity by Current-Wind Interaction.</i> Cho, A., Song, H., Moon, I.-J., Seo, H., Sun, R., Mazloff, M. R., Subramanian, A. C., Cornuelle, B. D., & Miller, A. J.	Seoul, Rep. of Korea <i>May. 2025</i>
AOGS Annual Meeting (poster) <i>The Changes in the States of Atmosphere and Ocean, and their Exchanges by Surface Current Coupling.</i>	Pyeongchang, Rep. of Korea <i>Jun. 2024</i>

Cho, A., Song, H., Seo, H., Sun, R., Subramanian, A. C., Cornuelle, B. D., Mazloff, M. R., & Miller, A. J.

KSO Spring Meeting (talk)

Seogwipo, Rep. of Korea

The Changes in the states of Atmosphere and Ocean, and their fluxes by surface current coupling

May. 2024

Cho, A., Song, H., Seo, H., Sun, R., Subramanian, A. C., Cornuelle, B. D., Mazloff, M. R., & Miller, A. J.

EGU General Assembly (talk)

Vienna, Austria

Atmospheric and Oceanic Responses to Surface Current Coupling near the Kuroshio Current.

Apr. 2023

Cho, A., Song, H., & Seo, H.

AGU Fall Meeting (talk)

Chicago, IL, USA

Mesoscale Dynamic and Thermodynamic Coupling between the Atmosphere and the Ocean near the Kuroshio Current.

Dec. 2022

Cho, A., Song, H., & Seo, H.

KMS Fall Meeting (talk)

Gwangju, Rep. of Korea

Atmosphere-driven cold SST biases over the western North Pacific in the GloSea5 seasonal forecast system.

Oct. 2022

Cho, A., Song, H.

KSO Spring Meeting (talk)

Busan, Rep. of Korea

Heat transport by typhoon in WRF-MITgcm coupled model.

Jul. 2020

Cho, A., Song, H., Sun, R., Subramanian, A. C., Cornuelle, B. D., & Mazloff, M. R.

Service

Peer Reviewer: *Journal of Marine Systems*

Technical Skills

Programming: Python, Fortran

Numerical model: Weather Research Forecasting model (WRF), MIT General Circulation Model (MITgcm), WAVEWATCH III

HPC: PBS job scheduler, parallel computing on HPC clusters

Tools: Linux shell scripting, LaTeX, Git